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Foreign Player Quota, Experience and Efficiency of Basketball Players

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Abstract

This study analyzes how the efficiency of basketball players depends on experience, player's nationality and team type. Using data for players in the Turkish Basketball League (TBL) from 1997 to 2010, we first show that efficiency of players initially increase with experience, but after a certain level decrease with it. Second, foreign players are on average more talented than native players. Third, the experience–efficiency link is not valid for foreign players indicating that more experience does not affect efficiency. Fourth, efficiency of foreign players in top-teams is no different than that of foreign players in regular (non-top) teams; whereas, native players in top-teams are more efficient than native players in other teams. Finally, in the presence of foreign player quota, top-teams hire the maximum number of foreign players allowed, while financially constrained regular teams can afford less foreign players. Consequently, under foreign player quota the quality of native players and the extent to which a team can pay for additional foreign players determine the team success.

KEYWORDS: efficiency, basketball, foreign player quota, experience, panel data

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“Basketball is a team sport where collective play determines the essence of an individual behavior on the court.”

Trninić et al. (2000), pp. 444.

1. Introduction

Individual talent is a necessary ingredient for a successful career in sports, although it is not sufficient by itself. What determines success is how effectively and efficiently an athlete combines his talent with mental skills in pursuit of constant improvement of game capacity. In basketball sense, although talent is a nature's gift, ability to fully internalize the philosophy of the game comes with practice and maturity. Hence, experience is a key factor for efficient use of individual talent in the name of collective success of the team.

In economic terms, efficiency is simply the comparison of what is actually produced with what can be achieved by consuming the same amount of resources. In sports, the resources available to an athlete are physiological aspects like energetic ability, flexibility and muscular strength; technical aspects like the precision in performing a fundamental movement and mental aspects like understanding and applying tactics. Hence, how good these resources are utilized up to their potential determines individual efficiency in sports. Although the set of abilities shaping up the efficiency is not directly observable, it is reflected in game statistics which can be considered as indicating implied efficiency. Therefore considering individual game statistics like points, assists, turnovers in basketball as available inputs, it is possible to come up with an efficiency measure. In its simplest form, efficiency of a basketball player can be defined as the sum of positive (such as points, assists, rebounds) and negative inputs (such as field goals missed) utilized for producing a win for the team.

Efficiency of a player is a reflection of his quality, and therefore, an important indicator of his market value as well. Considering skill abundance and comparative advantage along with international mobility, one may expect all the domestic leagues to be filled with players from a certain set of nations. However, existence of foreign player quotas pushes the demand for domestic players up along with their market value. In this respect it is of great value to determine how efficient foreign and native players are and whether foreign players are in fact more efficient than native ones. The answers to these questions have important economic implications for what type of players and from which source to hire under foreign player quotas.

Using data for players in the Turkish Basketball League (TBL) from 1997 to 2010, in this paper we first show that efficiency of players initially increase with experience, but after a certain level decrease with it. Second, we show that foreign players are on average more talented than native players. Third, the experience–efficiency link is not valid for foreign players indicating that more experience

does not affect their efficiency. Fourth, efficiency of foreign players in top-teams is statistically no different than that of foreign players in regular (non-top) teams; whereas, native players in top-teams are more efficient than native players in other teams. Finally, in the presence of foreign player quota, top-teams hire the maximum number of foreign players allowed, while financially constrained regular teams can afford less foreign players. Hence, a basketball team's success comes from the quality of its native players and additional foreign players it can sign under foreign player quota.

The next section presents an overview of the literature and methodology related to the empirical framework. In Section 3, data, model, estimation results and main findings are presented. Finally, section 4 concludes the paper.

2. Overview of the Literature and Methodology

2.1. An Overview of the Literature

In today's global environment many goods and services, including athletic talent, are traded between countries. Basketball is not an exception as players can move freely around the world to a large extent. Yet, national basketball organizations are allowed to establish quotas for the number of foreign players that domestic teams can contract. For example, Falcous and Maguire (2005) report that while most European leagues allowed only two foreign players, five players were allowed in Britain in late 1990s. Heterogeneity among the quota numbers of national organizations is still present in European basketball, even though quotas are being relaxed and the number of foreign players is increasing. Alvarez et al. (2011) report that after 1995, the average number of foreign basketball players in European leagues increased from 2.46 to 5.62 per team in 2007. As of 2010 in Russia, there is a requirement of seven domestic players per team and a minimum of one domestic player on the court during the play of the game. In Greece only two non-EU players are allowed. In Turkey five foreign players are allowed per team in a game roster, although at most three of them can play on the court at the same time. In Lithuania and Germany there are no restrictions on the use of foreign players.

The main argument behind the imposition of foreign player quota in sports is the possible negative spillover effects of foreign players on the development of local talent and on the performance of the national team. Baur and Lehmann (2007) report that as the number of imported players increases, the foreigners capture more of the game time and national players accumulate less experience. Verbon (2008), stressing the implications for domestic player development, shows that without any quota rules, small leagues will lose the incentive and the motivation to train local talents and the outcome will be the undersupply of local talent. On the other hand, Alvarez et al. (2011) find that increased number of

foreign players in domestic league enhances the performance of local basketball players in their respective national teams.

Protectionism through imposing a quota has implications on the demand for and the competitiveness of domestic players in this case, and on the quality of the domestic league in general. The existence of foreign player quotas increases the demand and the wages of native players. Lang, Rathke and Runkel (2009) investigate the effects of restrictions on foreign players with a model based on contract theory. They show that when the restriction on the number of foreign players is binding, the competitive balance, wages for local players and profits of the clubs are higher compared to the case where restrictions are not binding. However, Flores, Forrest and Tena (2010) show that liberalization could improve the competitive balance depending on the size of the talent pool. Overall, the direction of the spillover effect of foreign players is rather undecided in the literature.

In sports economics literature studies on the contribution of players to team level efficiency and productivity, it is typical to use a production function approach. Berri (1999) and Lee and Berri (2008) use individual, whereas Hofler and Payne (1997, 2006) use team level data for analyzing the productivity of National Basketball Association (NBA) teams. Other examples of the production function approach include Hadley et al. (2000) for National Football League (NFL) teams, Kahane (2005) for National Hockey League teams, Barros and Leach (2007) for English Premier League teams. Jane et al. (2010) analyze the effect of individual efficiency on club performance using data from professional baseball and show that the club level efficiency –accumulation of individual technical efficiencies- has an impact on club performance. Similarly, Teramoto and Cross (2010) examine the contributions of efficiency and related factors to winning games in both regular season and the playoffs in NBA. Bai (2009) examines how environmental factors affect the individual efficiency of players from NBA. These studies however do not make comparison between the efficiency of certain type of players. Specifically, in sports economics literature, what lacks is the comparison of efficiency of foreign and domestic players under the presence of foreign player quotas.

Another interesting issue to explore is how the level of experience affects efficiency and how this relation differs for foreign and native players when there is a foreign player quota. In the labor economics and population studies literature, there is a considerable amount of evidence on the link between age/experience and productivity. For instance, Skirbekk (2004), providing a survey, documents that the productivity profile of a working person as a function of age is initially low, then increasing, and later decreasing as the person's job related skills start to deteriorate. Skirbekk (2008), in addition, finds that the shape of this relation changes with the abilities demanded in the market. Marchetti (2002) shows that for professions requiring both physical and mental skills such as for athletes,

productivity peaks at an earlier age compared to professions requiring mostly mental skills. In the context of athletes, for instance, Tiruneh (2010) shows that the success of golfers in tournaments depends on age and that the age-success profile is hump-shaped.

In addition to player efficiency, financial strength is also an important factor for the success of a team. Generally, teams in European sports leagues are more concerned with maximizing success in winning rather than profit. Teams in small-sized leagues have difficulties in raising revenues from outside sources and are generally financed by locally-generated revenues and the league's governing body. Further, if the league is also open to promotion and relegation, then teams with unstable financing are generally relegated, if not they cease to exist. Höchtl and Konovalov (2007) show that when teams are dependent on local revenue and are price taker in the market for players, an unstable equilibrium may exist and financially inadequate teams may drop out following a shock. As an example, Turkish basketball league was played with only 12 teams in 2001-2002 season, due to the teams dropping out for financial reasons following an economic crisis. The fragile financial status of teams and their focus on winning may harm the size and quality of the league, and financially strong teams are more likely to succeed by hiring more talented native and foreign players.

2.2. Overview of the Methodology

This study analyzes how the efficiency of basketball players depends on experience, player nationality and team type. In order to analyze this one may use data for a single team in one season or for all the teams in one season, which yields a cross section data set. Instead, more generally one may use data for all the teams in all the seasons available, where in this case, we have a panel dataset including observations for a large number of players over many seasons. An advantage of the panel data is that it provides tools for handling the unobserved heterogeneity –the unobserved player specific factors affecting their efficiency in our context. Panel data can be modeled by use of fixed or random effects models. A fixed effects model assumes that the unobserved individual heterogeneity is correlated with covariates of the model and uses the within group estimator in order to deal with this type of unobserved effect. In contrast, random effects model assumes that the individual heterogeneity is random and the model is estimated by a feasible generalized least squares procedure. Baltagi (2008) provides an extensive presentation of the panel data estimation methods. In the case of basketball players, unobserved individual effect may be attributed to inherit ability or talent of a player, which is obviously correlated with covariates such as team membership or the number of playing minutes. In the case of a panel data with unobserved individual effects matter, the use of the fixed effects estimator more appropriate for basketball players, as reported in many studies including Lee and Berri (2008), Hofler and Payne (2006).

Fixed effects estimator can alternatively be obtained by the Least Squares Dummy Variable (LSDV) approach, where a dummy variable for each individual player is included in the estimation. In our context, the LSDV approach is more useful as the estimates of unobserved individual effects (individual talent) are also provided. Although the coefficients of covariates in the LSDV model are consistent, the coefficients of fixed effects may be inconsistent for small time periods. However, this methodology is still valid since we are interested only in the relative comparisons of unobserved talents of different types of players. Apart from the individual unobserved effects, team and season specific factors may also affect the efficiency-experience relation. Thus in order to control for these factors, dummy variables for each team and season are also included in the estimation procedure.

3. Empirical Analysis

3.1. Turkish Basketball League

Basketball is the second most followed sport in Turkey after football, yet it receives increasingly more public attention from year to year. Recently, in the last World Championship, 2010, the Turkish national basketball team marked an historical accomplishment by winning the silver medal. Currently, Turkish national men's team ranks 6th on the FIBA's (International Basketball Federation) latest rankings.

The Turkish Basketball League (TBL) was formed officially in 1966 and is managed by the Turkish Basketball Federation. Over this 44 year-period, more than 60 different teams competed in the league, while only 11 of them have won the league title. Efes Pilsen Spor Kulübü (Efes Pilsen), has recently dominated the league with 13 championships and won the Korac Cup in 1996, first European title won by a Turkish team at any team sports. In the last decade, Efes Pilsen, Ülkerspor and Fenerbahçe&Ülker are the only teams winning the title, while Beşiktaş, Galatasaray and Türk Telekom are among the top performers in the league. TBL currently ranks 6th in domestic league rankings of ULEB (Union of European Leagues of Basketball).

TBL is played with 16 teams. At the end of the regular season, the first 8 teams advance to play-offs, while the last two are relegated to the second division. The champion is determined after a best-of-seven final series. Although the play-offs determine the champion, the standings in regular season play an important role for the play-off match-ups, thus the regular season has a similar competitive level as the play-offs. For instance, in the last 12 seasons, the league title was won seven times by the first-place team, four times by the second-place team and only once by the third-place team of the regular season.

TBL teams are subject to a quota concerning the number of foreign players allowed in their game roster. This foreign player quota has been relaxed several

times in last two decades (Table 1). The quota mandates the number of foreign players in the team roster and the number of foreign players playing on the court during a game.

Table 1: Foreign Player Quota in the Turkish Basketball League (TBL)

Seasons	Number of foreign players in the game roster	Number of foreign players playing on the court
1997-1998	2	2
1998-1999 to 2001-2002	3	2
2002-2003 to 2006-2007	4	2
2007-2008 to 2010-2011	5	3

Notes: The first column refers to the maximum number of foreign players allowed in the game roster of 12 players. The second column refers to the maximum number of foreign players allowed among 5 players playing on the court during the game.

3.2. The Data

The data is retrieved from the official statistics of TBL, publicly available at <http://basketbol.tblstat.net>. Dataset includes the name, date of birth, team, number of games played and average efficiency information for all 1113 players who played in the TBL from 1997-1998 to 2009-2010 seasons. Dataset also contains season average figures for points, rebounds, assists and field goal shooting percentage for regular season games. The team, age and games played information are matched and each data row corresponds to a player at a season.

It is important to discuss how the efficiency is defined and how it differs from productivity. Individual productivity of a basketball player refers to the final outcomes like points scored, rebounds grabbed, assists made and shots blocked. Efficiency, on the other hand, is about how effectively a player produces this output. Using individual statistics, as a general practice for efficiency, observed positive outcomes are adjusted for turnovers and number of shots missed. This study also uses this generic definition of an efficiency index: $\text{Efficiency} = [\text{Points} + \text{Rebounds} + \text{Assists} + \text{Steals} + \text{Blocks} - (\text{Field Goals Attempted} - \text{Field Goals Made}) - (\text{Free Throws Attempted} - \text{Free Throws Made}) - \text{Turnovers}]$. This efficiency index is widely used officially by professional leagues including the NBA, TBL and many European basketball leagues as reported in Martínez (2010). Moreover, this index is also used in various studies such as Coates and Oguntimein (2010), Groothuis, Hill and Perri (2009), Rice and Sen (2008) among others. Descriptive statistics of all the variables in the dataset are presented in Table 2:

Table 2: Descriptive Statistics of Variables

	Obs.	Mean	Std. Dev.	Min.	Max.
Efficiency	2705	7.50	5.88	-4	28.08
Age	2705	24.60	4.54	15	39
Age-squared	2705	625.89	227.47	225	1521
Games played	2705	18.19	9.38	1	30
Foreign	2705	0.32	0.47	0	1
<i>other (season averages)</i>					
Points	2705	7.29	5.57	0	28.21
Rebounds	2705	3.08	2.51	0	13.27
Assists	2705	1.26	1.24	0	8.76
<i>shooting percentage</i>					
Free throw	2398	65.82	17.42	0	100
Two-point field goals	2563	48.62	15.42	0	100
Three-point field goals	2300	30.15	16.84	0	100

Among 35 different teams in the sample, six of them stand out as *top-teams* in TBL in terms of success, financial strength and popularity. These are Efes Pilsen, Ülkerspor, Fenerbahçe (Fenerbahçe&Ülker after 2006 as the two teams merged), Beşiktaş, Galatasaray and Türk Telekom. These teams have the highest average rankings in standings in the seasons covered, and they are the only ones reaching the top-three position of the standings. Another team, Tofaş, has reached the top-three position twice, but that was in the first three years of the sample, after which the team withdrew from the league. Moreover, Beşiktaş, Fenerbahçe and Galatasaray are the biggest sport clubs in Turkey with millions of supporters. In terms of nationality, out of 1113 players, 493 are Turkish and 620 are foreign. The descriptive statistics of efficiency and age of players by nationality and team type are reported in Table 3.

Table 3: Efficiency and Age by Nationality and Team Category

Top-teams					
	Efficiency			Age	
	Obs.	Mean	Std. Dev.	Mean	Std. Dev.
Foreign players	365	12.96	4.97	27.30	3.39
Turkish players	678	5.85	4.58	23.94	4.64
Regular (Non-top) teams					
	Efficiency			Age	
	Obs.	Mean	Std. Dev.	Mean	Std. Dev.
Foreign players	497	12.93	5.42	26.48	3.38
Turkish players	1165	4.43	3.85	23.35	4.62

A general property of the efficiency index should be addressed. Efficiency is a cumulative index and depends on the number of minutes a player plays in a game. The relationship between efficiency and minutes played is not clear. Initially, playing more minutes may increase the efficiency, but after a threshold, it may also decrease the efficiency if the player gets tired and accumulates mistakes. Sampaio et al. (2010) show that even though the playing time matters, errors made is the most important factor distinguishing important and less important basketball players, in other words, efficient and less efficient players. Moreover, Berri and Bradbury (2010) document that the correlation between NBA efficiency index (not corrected for minutes) and Game Score index (an efficiency index corrected for minutes) is 0.99 for a sample of NBA players from 2007 to 2009. Therefore, in this study we will focus on the generic efficiency index, unadjusted for playing time, as described above.

3.3. Model

In this study, we consider a model where player efficiency is a function of experience and number of games played in a season. Efficiency is expected to increase first and later to decrease with experience after a certain level, where age of a player is used as a proxy for experience. To capture the nonlinear relationship between efficiency and age, the variables age and age-squared are included in the model. Using age as a proxy for experience seems to be a fair assumption in the case of basketball players, as most players start their careers at a similar age in early school years. Bai (2009) also follows this approach. The number of games a player played in a season is included in the model as well, since playing practice also affects efficiency.

A player's efficiency depends on the team and season in which he plays, as well as his talent. Efficiency of a player is also related to the performance of teammates. For example, one player who takes many shots may have a negative impact on other players' efficiencies, and a player's efficiency may be enhanced on a team with better players. Idson and Kahane (2000) report that in National Hockey League team characteristics affect the performance of individual players; while Torgler and Schmidt (2007) show that teammate characteristics affect the individual performance in soccer. To obtain unbiased estimates of age affects, one needs to control for other effects by including individual, team and season-specific fixed effects in the regression model. To include fixed effects:

For players $i=1, \dots, N$; $N-1$ dummy variables, $PD_2_i, PD_3_i, \dots, PD_N_i$, are created, where $PD_2_i = \begin{cases} 1 & \text{for } i = 2 \\ 0 & \text{otherwise} \end{cases}$, etc.

For seasons $t=1, \dots, S$; $S-1$ dummy variables, $SD_2_t, SD_3_t, \dots, SD_S_t$, are created, where $SD_2_t = \begin{cases} 1 & \text{for } t = 2 \\ 0 & \text{otherwise} \end{cases}$, etc.

For teams, $j=1, \dots, T$; T-1 dummy variables, $TD_2_j, TD_3_j, \dots, TD_T_j$, are created, where $TD_2_j = \begin{cases} 1 & \text{for } j = 2 \\ 0 & \text{otherwise} \end{cases}$, etc.

Let $Eff_{i,j,t}$ denote the average efficiency of the i^{th} player in the j^{th} team and t^{th} season. The regression equation for the player efficiency including player, team and season fixed effects is given by:

$$\begin{aligned} Eff_{i,j,t} = & \alpha + \beta_1 Age_{i,t} + \beta_2 Age_{i,t}^2 + \beta_3 G_{i,t} + (\tau_2 PD_2_i + \dots + \tau_N PD_N_i) \\ & + (\delta_2 SD_2_t + \dots + \delta_S SD_S_t) + (\gamma_2 TD_2_j + \dots + \gamma_T TD_T_j) \\ & + (\pi_2 TSD_2_q + \dots + \pi_{S*T} TSD_S * T_q) + \varepsilon_{i,j,t} \end{aligned} \quad (1)$$

For N players over S seasons and T teams, the above equation is estimated, where G is the games played; and PD_(2 to N), SD_(2 to S), TD_(2 to T) and TSD_(2 to S*T) refer to player, season, team and team by season dummy variables respectively. The error term is assumed to be normally distributed with 0 mean and a constant variance.

3.4. Estimation Results

The LSDV estimation results of equation (1) are presented in Table 4. The table reports the coefficient estimates of age and age-squared on player efficiency after controlling for games played and individual, team and season-specific factors. The first column reports the coefficient estimates for the sample of all players, while the second and third columns report the estimates for samples of foreign and native players respectively. When analyzing the estimation results, the significance criteria for the coefficients is the 5% level as it is the generally accepted level in social sciences. For the sample of all players, the coefficients of age, age-squared and games played are statistically significant even at 1% level. It is also the case for the sample of native players. For the sample of foreign players, however, coefficients of age and age-squared are not statistically significant at 5% level.

Based on these results, the analysis of this paper reveals the following: First, considering the sample of all players, efficiency of a player increases with experience up to a certain age after which efficiency starts to decline. This nonlinear nature of relationship between efficiency and experience is evident from the estimation results; where the coefficient of age is statistically significant and positive while the coefficient of age-squared is statistically significant and negative (Table 4, Column 1).

Table 4: LSDV Regression of Efficiency on Experience

Dependent variable:	Efficiency		
	(1)	(2)	(3)
Age	2.8206 [0.2429] (0.000)	5.0656 [4.0335] (0.211)	1.8427 [0.2422] (0.000)
Age-squared	-0.0464 [0.0048] (0.000)	-0.0870 [0.0698] (0.214)	-0.0368 [0.0047] (0.000)
Games played	0.1455 [0.0153] (0.000)	0.2033 [0.0926] (0.029)	0.1426 [0.0158] (0.000)
Constant	-41.7127 [4.7506] (0.000)	-45.2943 [56.1460] (0.421)	-17.9142 [3.1153] (0.000)
Sample	All players	Foreign players	Native players
Number of obs.	2705	862	1843
R ²	0.90	0.97	0.79

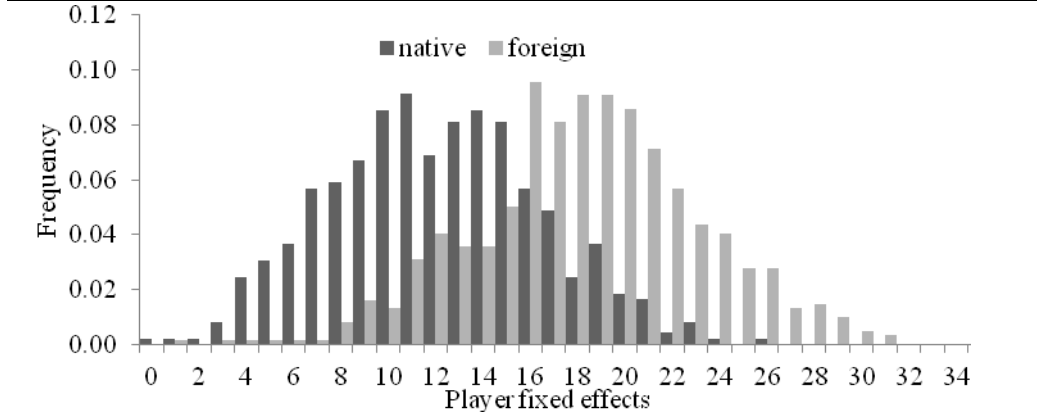
Notes: Standard errors -in squared brackets- are clustered at team by season level. P-values are reported in parenthesis and in italics. Regression model includes individual, season, team and team by season specific dummy variables. A p-value>0.05 indicates that the coefficient is not statistically significant at 5% level.

The coefficients of the dummy variables for each player in the LSDV regression give the estimates of individual specific effects. These player specific effects include factors such as talent, personal motivation, coach's preferences, minutes played per game. Yet, here we assume that the individual fixed effects reflect the unobserved individual talent. With this, a visual exploration of the distribution of individual fixed effects suggests that foreign players are more talented than native players on average (Figure 1).

Second, the experience–efficiency link is not valid for foreign players as the coefficients of age and age-squared are not statistically significant at 5% level (Table 4, Column 2). That is, for foreign players additional experience has no significant effect on efficiency. In addition, once the regression model is re-run with the average points scored and the two-point field goal percentages replacing efficiency as the independent variable, the coefficients of age and age-squared are still not statistically significant. This indicates that neither the points nor the field goal percentages of foreign players change with experience. On the other hand, the effect of experience on efficiency is statistically significant for Turkish players (Table 4, Column 3).

Given that foreign players perform almost equally efficiently whether more experienced or not and considering that experienced foreign players are more costly to employ, this suggests that teams with limited budgets could maximize their overall efficiency by employing younger and less costly foreign players.

Figure 1. Distribution of Estimated Individual Fixed Effects



Notes: Individual fixed effects are the coefficients of individual player dummies in the estimation. The individual specific effects are assumed to be the indicator of the unobserved talent of a player. Hence, the higher the estimated individual fixed effect, the more talented a player is. The lowest individual specific effect is normalized to 0. The relative frequencies within each group are shown on the vertical axis. There are 493 native and 620 foreign players.

As mentioned in the data section, the efficiency index does not consider the minutes played in a game. In order to check whether adjusting for minutes changes the results, we performed the following analysis. In the dataset, minutes played per game information is available for the 2006-2007 season. To make use of this information, we divide the players into four groups: foreign players in top-teams; foreign players in regular (non-top) teams; Turkish players in top-teams and Turkish players in regular (non-top) teams. Then we calculate the average minutes played per game for each type of players. Using the information from this season, we adjust the efficiency figures of all the players for all the seasons to 40 minutes according to the group they belong to. Finally, we estimate the equation (1) using the adjusted efficiency as the dependent variable. The results reveal that the sign and the significance level of the coefficients are the same as the ones reported in Table 4. In other words, adjusting efficiency for minutes played does not change the conclusions drawn.

Next we explore if the efficiency of foreign players playing in top-teams is different from the efficiency of foreign players in other teams. In order to see this, introducing a “top-team” dummy variable and controlling for the number of games played together with season and team fixed effects, the following specification is estimated:

$$\begin{aligned}
 Eff_{i,j,t} = & \alpha + \beta_1 Toplevel_{i,t} + \beta_2 G_{i,t} + (\delta_2 SD_2_t + \dots + \delta_S SD_S_t) \\
 & + (\gamma_2 TD_2_j + \dots + \gamma_T TD_T_j) \\
 & + (\pi_2 TSD_2_q + \dots + \pi_{S*T} TSD_S * T_q) + \varepsilon_{i,j,t}
 \end{aligned} \tag{2}$$

where $Topteam_{i,t} = 1$ if the i^{th} player played in a top-team in t^{th} season and 0 otherwise; and $TD_{(2\ to\ T)}$, $SD_{(2\ to\ S)}$ and $TSD_{(2\ to\ S*T)}$ refer to team, season and team by season dummy variables. The error term is assumed to be normally distributed with 0 mean and a constant variance. The estimation results from this specification are reported in Table 5.

Table 5: OLS Regression of Efficiency on Top-Team Dummy		
Dependent variable:	Efficiency	
	(1)	(2)
Top-team	-1.1779 [0.8816] (0.183)	1.3783 [0.5122] (0.008)
Games played	0.2825 [0.0232] (0.000)	0.2836 [0.0088] (0.000)
Constant	5.3228 [0.4000] (0.000)	-8.7418 [0.7155] (0.000)
Sample	Foreign players	Native players
Number of obs.	862	1843
R ²	0.46	0.51

Notes: Standard errors -in squared brackets- are clustered at team by season level. P-values are reported in parenthesis and in italics. Regression model includes season, team and team by season specific dummy variables. A p-value>0.05 indicates that the coefficient is not statistically significant at 5% level.

The results show that the top-team dummy variable is not statistically significant for the sample of foreign players (Table 5, Column 1). This implies that foreign players playing in top-teams are not more efficient than foreign players playing in regular (non-top) teams on average.

Based on the findings of the two sets of estimations, we argue that two factors stand out in determining team success in the TBL, under the presence of foreign player quota, where success is associated with being a top-team in our context. These factors are the quality and efficiency of the native players, and the additional foreign players top teams can afford to employ. First, native players in top teams are more efficient than native players in regular (non-top) teams (Table 5, Column 2). Second, while top teams enjoy the maximum number of foreign players allowed in their roster, regular teams can afford less foreign players on average, as they are financially more constrained and foreign players are more costly than native counterparts. In 2006-2007 season, where detailed game data is available, when the foreign player quota was 4, top-teams averaged 3.7 foreign players and regular teams averaged 2.9 foreign players on their rosters. Detailed game data is also partially available for the current 2010-2011 season where the foreign player quota is 5. In the current season top-teams averaged 4.9 foreign players and regular teams averaged 3.7 foreign players on their rosters. These

figures indicate that regular (non-top) teams usually hire one foreign player less than top-teams on average. This observation points to an additional mechanism for financial strength to affect team success, given that on average foreign players are more efficient than native players.

4. Concluding Remarks

The limitations on international mobility of players via foreign player quotas have important implications in the context of the study. First, it increases the demand for native players, even though they are not necessarily as efficient as foreign players. As foreign players are mobile, it is easy to sign a new player in place of an inefficient one, and therefore the turnover rate of foreign players is high. On the other hand, as the market for native players is quite tight, it is not easy to replace an inefficient player, and eventually, there is always demand for less efficient players. Second, regular (non-top) teams need to spend more resources on domestic player development for survival given that native players must be used and experienced natives are expensive to employ. Third, the finding that foreign players perform almost equally efficiently in top and regular (non-top) teams implies that regular teams are successful at recruiting foreign players that could match the efficiency of foreign players playing at top-teams. This research suggests additional questions for future study. The effects of the changes in foreign player quota on the player choice of teams and whether foreign players harm the domestic player development should be analyzed with more detailed data. Another question for future study is how to strengthen the financial status of teams to improve their sustainability and the competitive balance of the league.

Appendix: Variable Definitions

Efficiency	: Average efficiency in a given season
Efficiency per 40 min.	: Average efficiency adjusted for average playing time
Age	: Age of a player in a given season
Age-squared	: Squared age of a player in a given season
Games played	: Number of games a player played in a given season
Top-team	: Dummy variable, =1 if team is a top team; =0 o/w
Points	: Average points scored in a given season
Two-point shooting %	: Average two-point field goal % in a given season

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